

Product Listing Generator – Brief Report

How the API Integration Works

This project uses the OpenAI API to generate product listings based on product images. Images are read from a local folder and converted into a format that can be sent to the API together with a textual prompt. For each product image, a request is sent to the model asking it to analyze the visual content and generate a structured product description, including key attributes such as title, category, condition, features, and a short description. The integration is designed to process multiple products in sequence, allowing the same logic to be applied consistently across different images. The API response is handled programmatically and prepared for further use within the application.

Challenges Faced

One of the main challenges was ensuring consistent and structured responses from the API, especially when working with image-based inputs. Managing local file paths and making sure images were correctly loaded and passed to the API also required careful handling. Additionally, working with external APIs requires attention to request formatting, response handling, and error management, which added complexity to the implementation. These challenges were addressed through iterative testing and incremental improvements to the code structure.

Quality of Generated Listings

The generated product listings are generally clear and relevant to the products shown in the images. The model is able to identify product categories and generate readable descriptions that resemble typical e-commerce listings. Key features are presented in a concise and understandable way, making the output suitable for practical use cases such as online catalogs or marketplaces.

Potential Improvements

There are several potential improvements for future development. Integrating additional product metadata could help improve accuracy and reduce ambiguity in the generated listings. Another improvement would be to refine prompts and validation logic to further enhance output consistency. Finally, optimizing the processing workflow and adding more robust error handling would make the system more scalable and reliable when working with larger datasets.